Learning from Forecast Errors

Methods and Toolkit

Forecasting in central banks, CCBS online seminar

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Methods based on: Macro Technical Paper No. 6

Python toolkit: github.com/bank-of-england/forecast_evaluation

Demo for this course: github.com/paullabonne/forecast_eval_teachin

Agenda

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A quick primer on Python

Environment & packages — Python uses *virtual environments* to keep project dependencies separate. You install packages with `pip`:

```
pip install forecast_evaluation
```

Importing — load a package into your script, optionally with a short alias:

```
import forecast_evaluation as fe   # now use fe.something()
```

Classes & instances — a *class* is a blueprint; an *instance* is a concrete copy built from it. `ForecastData` is the blueprint; `data` below is your instance, holding your specific forecasts:

```
data = fe.ForecastData(your_outturns, your forecasts)  # create an instance
data.forecasts            # attribute: the data it holds
data.add_benchmarks(...)  # method: a function that works only on the instance
```

Storing macroeconomic forecasts

Each row answers: *"What value did source **S** assign to variable **Y** for period **t**, as of vintage **v**?"*

Column	What it means	Examples
`variable`	What is being forecast	`cpi`, `gdpkp`
`metric`	Transformation applied	`yoy`, `pop`, `levels`
`frequency`	Data frequency	`Q`, `M`
`date`	*Which period* the value refers to	`2022-03-31` = 2022Q1
`vintage_date`	*When* the number was known	`2021-09-30` = as of 2021Q3
`source`	Who produced it	`AR(p)`, `BVAR`, `SPF`
`forecast_horizon`	Periods between vintage and date	`+4` = 4Q ahead; `-k` = revision
`value`	The number itself	`0.064`

Storing macroeconomic forecasts

`your_forecasts`:

```
        date vintage_date variable  source frequency metric  forecast_horizon   value
  2015-03-31   2015-03-31    cpi   ar(p)         Q    pop                 0  0.0110
  2015-06-30   2015-03-31    cpi   ar(p)         Q    pop                 1  0.0150
  2015-03-31   2015-03-31    cpi    bvar         Q    pop                 0  0.0120
  2015-06-30   2015-03-31    cpi    bvar         Q    pop                 1  0.0160
```

`your_outturns`:

```
        date vintage_date variable frequency metric  forecast_horizon   value
  2015-03-31   2015-06-30    cpi         Q    pop                -1  0.0108
  2015-03-31   2015-09-30    cpi         Q    pop                -2  0.0112
  2015-06-30   2015-09-30    cpi         Q    pop                -1  0.0198
  2015-06-30   2015-12-31    cpi         Q    pop                -2  0.0203
```

```python
import forecast_evaluation as fe

data = fe.ForecastData(
    forecasts_data=your_forecasts,
    outturns_data=your_outturns,
)
```

Three dimensions of forecast quality

Dimension	Question	Key methods
Accuracy	How close are forecasts to outcomes?	RMSE, MAE, Diebold-Mariano
Bias	Do forecasts systematically over- or under-predict?	Mean error test, Mincer-Zarnowitz
Efficiency	Is all available information used optimally?	Nordhaus (weak), Blanchard-Leigh (strong)

These are **complementary** — a forecast can be accurate on average but still biased or inefficient.

The forecast error

The h-quarter-ahead error for variable y:

$$\varepsilon(y; k)_{t|t-h} := y_{t|t+1+k} - \hat{y}_{t|t-h}$$

where the outturn vintage k controls which data release is used as the "truth". We often set $k = 12$ (~3 years after the reference quarter).

Serial correlation at horizons $h > 1$: even under an optimal forecast, h-step-ahead errors follow an MA$(h-1)$ process.

Standard errors in tests must therefore be HAC-robust (e.g. Newey-West with at least $h-1$ lags).

Some sniff tests

```
# Hedgehog: each line = one vintage; dark line = outturns
fe.plot_hedgehog(
    data=data, variable="cpisa",
    forecast_source="mpr", metric="yoy",
    frequency="Q", k=12,
)

# Errors over time (with moving-average smoothing)
fe.plot_errors_across_time(
    data=data, variable="cpisa",
    metric="yoy", horizons=4, sources="mpr",
    frequency="Q", k=12, ma_window=4,
)
```

- **Hedgehog chart**: overlay all forecast vintages against outturns — reveals persistent over/under-shooting
- **Errors over time**: spot structural breaks, outlier episodes, trending errors

Accuracy: how large are the errors?

$$\mathrm{RMSE}_h = \sqrt{\frac{1}{N}\sum \varepsilon_{i,h}^2} \qquad \mathrm{MAE}_h = \frac{1}{N}\sum |\varepsilon_{i,h}|$$

Both should monotonically grow with horizon.

- **RMSE** penalises large errors more; sensitive to outliers
- **MAE** treats all errors equally; more robust to outliers

```
accuracy = fe.compute_accuracy_statistics(data=data, k=12)

accuracy.plot(
    variable="cpisa", metric="yoy",
    frequency="Q", statistic="rmse",
)
```

Benchmark models

A **reference point** estimated with real-time vintages.

Random Walk: $\hat{y}_{t+h|t} = y_t$

AR(p): $y_t = \mu + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t, \quad \varepsilon_t \sim t(\nu, 0, \sigma)$

- Lag order $p \leq 2$ selected by BIC; stationarity enforced
- **Student-t errors** — heavy tails prevent large shocks (2008, 2020) from distorting estimation

```
data.add_benchmarks(
    models=["AR", "random_walk"],
    metric="pop",
)
```

Relative accuracy: beating the benchmark

$$\text{RMSE ratio}_h = \frac{\text{RMSE}_h^{\text{model}}}{\text{RMSE}_h^{\text{benchmark}}} \quad \begin{cases} < 1 & \text{model wins} \\ > 1 & \text{benchmark wins} \end{cases}$$

The **Diebold-Mariano test** asks if the difference is *significant*.
Define $d_t = \varepsilon_{t,h}^{A\,2} - \varepsilon_{t,h}^{B\,2}$ and test $H_0 : \mathbb{E}[d_t] = 0$ with HAC standard errors. Harvey et al. (1997) small-sample correction applied.

```
comparison = fe.compare_to_benchmark(
    df=accuracy.to_df(),
    benchmark_model="baseline ar(p) model",
    statistic="rmse",
)

dm = fe.diebold_mariano_table(
    data=data,
    benchmark_model="baseline ar(p) model",
)
```

Bias: mean error test

$$\varepsilon_{t,h} = \beta + u_t, \quad H_0 : \beta = 0$$

$\hat{\beta} > 0$: forecasts **underestimate** outturns; $\hat{\beta} < 0$: **overestimate**.
OLS with HAC standard errors (max lag $= h$).

Mincer-Zarnowitz — joint test of no level or slope bias:

$$y_{t+h} = \beta_0 + \beta_1 \hat{y}_{t+h|t} + u_{t+h}, \quad H_0 : \beta_0 = 0,\ \beta_1 = 1$$

```
bias = fe.bias_analysis(data=data, source="mpr", k=12)

bias.plot(
    variable="cpisa", metric="yoy",
    frequency="Q",
)
```

Efficiency: weak vs strong

Weak efficiency — did the forecaster use their *own past forecasts*?
If today's revision is predictable from last quarter's, information was incorporated too slowly.
Clean identification: the forecaster certainly had their own past numbers.

Strong efficiency — did the forecaster use *all available information*?
If errors in y are predictable from anything the forecaster knew, the forecast is inefficient.
Harder to test: requires knowing what was in the forecaster's information set.

Blanchard-Leigh bridges the two: it uses the forecaster's *own forecast* of another variable x. Since they produced $\hat{x}$ themselves, they certainly had it — so it inherits the clean identification of weak efficiency while testing whether the *pass-through* from x to y was correctly specified.

Efficiency: weak (Nordhaus 1987)

A forecast is **weakly efficient** if past revisions cannot predict future revisions.

$$R(y)_{t|t} = \alpha + \sum_{i=1}^{N} \beta_i R(y)_{t|t-i} + u_t, \quad H_0 : \beta_1 = \cdots = \beta_N = 0$$

Rejection signals **information smoothing**: news incorporated gradually.

```
we = fe.weak_efficiency_analysis(
    data=data, source="mpr", k=12,
)

print(we.to_df())
```

Efficiency: strong (Blanchard-Leigh)

Wald ratio $\omega = \beta/\delta$ measures misspecified pass-through:

$$\varepsilon(y)_{t+h|t} = \alpha + \beta\hat{x}_{t+j|t} + u \qquad x_{t+j} = \gamma + \delta\hat{x}_{t+j|t} + e$$

$\omega > 0$: pass-through underestimated; $\omega < 0$: overestimated; $\omega = 0$: efficient.

```
bl = fe.blanchard_leigh_horizon_analysis(
    data=data, source="mpr",
    outcome_variable="cpisa", outcome_metric="yoy",
    instrument_variable="gdpkp", instrument_metric="yoy",
)

bl.plot()
```

Unstable environments: the stationarity problem

All previous tests require that the **object of interest is covariance-stationary** — its mean and autocovariance do not change over time. What that object is depends on the test: the forecast error for bias tests, the loss differential d_t for Diebold-Mariano, revision sequences for weak efficiency, etc.

In practice this often fails:

- **Structural breaks** — policy regime changes, financial crises, pandemics shift the distribution
- **Evolving models** — forecasting frameworks are updated, changing error dynamics
- **Time-varying volatility** — the Great Moderation, post-COVID inflation

A full-sample test averages over all regimes. A bias that appeared after 2020 is diluted by 15 years of unbiased forecasts — the test says "no problem" when the *current* forecast process is broken.

Unstable environments: rolling and fluctuation tests

Rolling window (W observations): re-run the test on every consecutive sub-sample. Reveals *when* a problem emerged or disappeared.

Fluctuation test (Giacomini & Rossi, 2010): same rolling window, but with critical values adjusted for the multiple-testing nature of scanning across windows. The null is that the test is never rejected in any window.

```
rolling_bias = fe.rolling_analysis(
    data=data, window_size=40,
    analysis_func=fe.bias_analysis,
    analysis_args={"k": 12},
)
rolling_bias.plot(variable="cpisa", horizons=[0, 4, 8])

fluct_bias = fe.fluctuation_tests(
    data=data, window_size=40,
    test_func=fe.bias_analysis,
    test_args={"k": 12},
)
```

The dashboard

All the functionalities of the package can be accessed through a built-in dashboard.

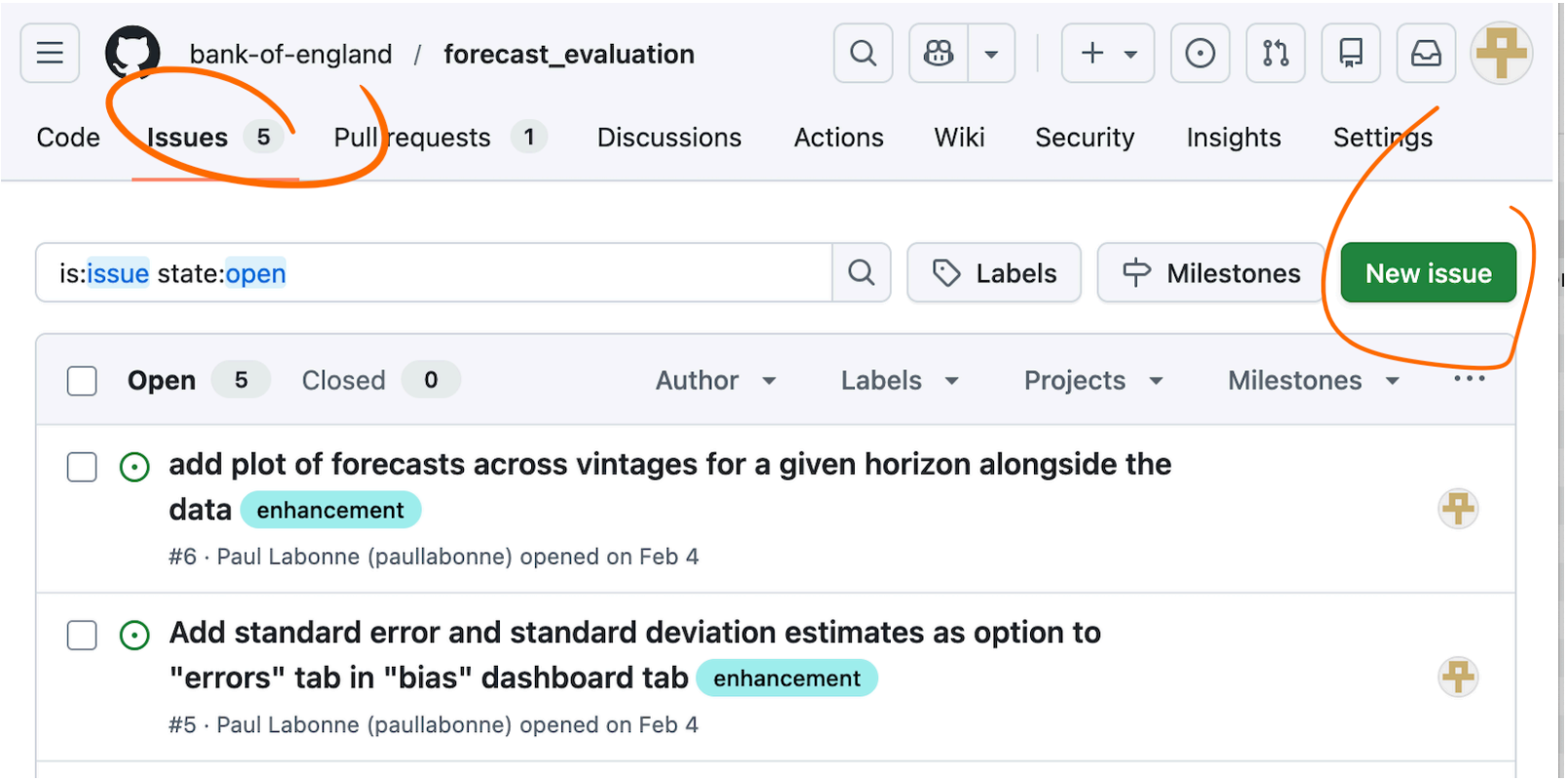
```
import forecast_evaluation as fe

data = fe.ForecastData(
    forecasts_data=your_forecasts,
    outturns_data=your_outturns,
)

data.run_dashboard()                # opens in browser
data.run_dashboard(from_jupyter=True)  # embed in Jupyter
```

Live demo

You can find the code for the demo here:

github.com/paullabonne/forecast_eval_teachin/demo.py

Have question? Want something? Found a bug?



`Thank you!`